9.3 Similar Right Triangles

Name _____ Period ____ Date _____

Geometry 9.3 · Student sheet

I can…

- show that the altitude to the hypotenuse makes three similar right triangles.
- find the geometric mean of two numbers.
- use the Geometric Mean Theorems to find missing lengths.

Vocabulary · fill in the missing word

Altitude to the hypotenuse — the segment from the right angle _____ to the hypotenuse

Geometric mean — of two _____ numbers a and b: the positive number x with $x^2 = ab$, so $x = \sqrt{ab}$

Similar triangles — triangles whose _____ angles are congruent and whose corresponding sides are proportional

Part 1 · Discovery: three triangles from one

Cut your card on a diagonal. Draw the altitude from the right angle to the hypotenuse and cut along it. Measure the angles of all three triangles.

Triangle	Right angle	Smaller acute angle	Larger acute angle
Whole triangle	90°		
Piece 1	90°		
Piece 2	90°		

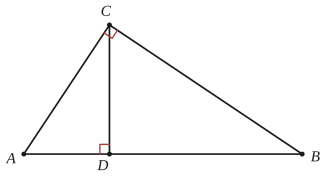
The three triangles are similar because _____.

Part 2 · Notes and examples (fill in each note, then work the example)

Right Triangle Similarity Theorem: the altitude to the hypotenuse of a right triangle forms two triangles that are _____ to the original triangle and to each other.

Example 1

In right △ABC, $\overline{CD}$ is the altitude to the hypotenuse. Name the three similar triangles.



Geometric mean of a and b: x = _____

Example 2

Find the geometric mean of each pair of numbers.

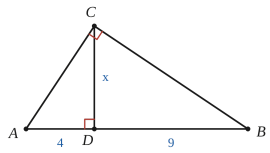
a. 4 and 25

b. 6 and 15

Geometric Mean (Altitude) Theorem: the altitude to the hypotenuse is the geometric mean of the two _____ of the hypotenuse: altitude^2 = (one segment)(other segment).

Example 3

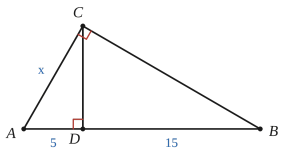
The altitude to the hypotenuse divides it into segments of 4 and 9. Find the length of the altitude, x.



Geometric Mean (Leg) Theorem: each leg is the geometric mean of the _____ and the segment of the hypotenuse _____ that leg: leg² = (hypotenuse)(segment next to the leg).

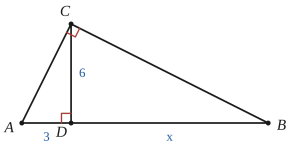
Example 4

The altitude to the hypotenuse divides it into segments of 5 and 15. Find the length of the leg next to the segment of 5, x.



Example 5

The altitude to the hypotenuse is 6, and one segment of the hypotenuse is 3. Find the other segment, x.



Part 3 · Partner practice

Partner 1

Find the geometric mean of 8 and 18.

Partner 2

The altitude to the hypotenuse divides it into segments of 2 and 8. Find the altitude, and the leg next to the segment of 8.

